

Annapurna Properties — AI Calling & CRM Workflow Report

Client Architectural & Workflow Specification Document | Date: September 2026

1. Executive Summary

This document outlines the end-to-end technical architecture, call lifecycle, integration flow, and database storage specifications for the **Annapurna Properties AI Calling & Real Estate CRM System**. The platform combines **Exotel Virtual Telephony IVR**, **ElevenLabs Speech-to-Speech AI**, **OpenAI GPT LLM Intelligence**, and **Meta WhatsApp Cloud API** into a unified **Django REST + PostgreSQL** CRM backend.

Component	Provider / Tech	Primary Function
Virtual Telephony	Exotel IVR Webhooks	Handles incoming customer calls & status callbacks
Voice Speech AI	ElevenLabs ConvAI	Real-time bidirectional speech-to-speech AI in Hindi/English
Intelligence Engine	OpenAI GPT LLM	Post-call intent extraction, summary & lead qualification
Messaging Nurture	Meta WhatsApp Cloud API	Sends instant WhatsApp confirmation templates to qualified leads
CRM Core & Database	Django REST + PostgreSQL	Manages Users, Leads, Calls, Transcripts, Summaries & Followups

2. End-to-End Call Lifecycle & Data Flow

Step 1: Incoming Call Initiation (Exotel Telephony)

When a customer calls the Annapurna Properties virtual number, Exotel sends a Webhook (POST `/api/v1/telephony/webhooks/incoming-call/`) with `CallSid`, customer phone number (e.g. `+919876543210`), and call direction. The system instantly creates a record in the `calls_call` table with status `Queued` and logs a `CallEvent`.

Step 2: Live AI Voice Conversation (ElevenLabs ConvAI)

The backend connects Exotel's audio stream with ElevenLabs Conversational AI WebSockets. The AI Agent conducts a natural speech-to-speech conversation in Hindi/English, qualifying customer requirements for property type (2BHK/3BHK), location (e.g. Sector 62), budget range (e.g. 1-1.5Cr), and site visit availability.

Step 3: Call Termination Callback (Exotel Webhooks)

When the call ends, Exotel sends a final status callback (POST `/api/v1/telephony/webhooks/call-status/`) containing duration (e.g., 140s) and status (`Completed`). The system updates the `calls_call` table status to `Completed` and launches the async Celery post-call processing pipeline.

Step 4: AI Post-Call Intelligence & Lead Qualification (OpenAI LLM)

The async pipeline executes three automated actions:

- **Transcript Log:** Fetches turn-by-turn conversation turns from ElevenLabs and saves them into calls_calltranscript.
- **LLM Intent Extraction:** OpenAI GPT processes the transcript, extracts structured parameters (Requirement, Budget, Location, Interest Level), and saves the summary into calls_callsummary.
- **Auto Lead Pipeline:** Creates a new row in leads_lead (or updates an existing lead), assigning status Qualified if interest is high.

Step 5: Instant WhatsApp Confirmation (Meta WhatsApp Cloud API)

For High/Medium interest leads, Celery triggers an automated WhatsApp message via Meta Graph API using the enquiry_ack_v1 template. Details are stored in whatsapp_whatsappmessage and a follow-up task is scheduled in followups_followup for the CRM relationship manager.

Step 6: Realtime CRM Agent Dashboard

The CRM Agent logs in via JWT authentication to view real-time metrics (/api/v1/dashboard/metrics/), review qualified leads, play call recordings, read AI summaries, and track WhatsApp delivery status.

3. PostgreSQL Database Table Specifications

Table Name	Key Fields & Data Stored
accounts_user	Stores Admin and CRM Agents, roles (admin / CRM), email & password hashes
accounts_password_reset_otp	Tracks 6-digit email OTPs, 15-min expiry timestamp, and usage status (is_used)
leads_lead	Lead profile, display ID (LD-1001), mobile, property requirement, budget, status, assigned agent
calls_call	Call session record, provider call SID, duration, direction (inbound/outbound), status (Completed)
calls_calltranscript	Turn-by-turn conversation dialogue (speaker: agent/user, text, turn order)
calls_callsummary	AI structured analysis: requirement, location, budget range, interest level, and paragraph summary
whatsapp_whatsappmessage	WhatsApp outbound log, message SID, delivery status (Sent/Delivered/Failed)
followups_followup	CRM Agent task assignment, due date/time, follow-up type, and completion status

4. Complete API Endpoint Reference (18 Endpoints)

Category	Method & Path	Auth Required	Description
Auth	POST /api/v1/auth/signup/	Public	Register new user/agent & get JWT tokens
Auth	POST /api/v1/auth/login/	Public	Email/username & password login
Auth	POST /api/v1/auth/reset-password/	Public	Request 6-digit email OTP for password reset
Auth	POST /api/v1/auth/reset-password/confirm/	Public	Confirm password reset using 6-digit OTP
Auth	POST /api/v1/auth/change-password/	JWT Bearer	Change existing password
Auth	POST /api/v1/auth/logout/	JWT Bearer	Blacklist JWT refresh token
Leads	GET /api/v1/leads/	JWT Bearer	List & filter CRM leads by status/search
Leads	POST /api/v1/leads/	JWT Bearer	Create manual lead profile
Leads	GET /api/v1/leads/{id}/	JWT Bearer	Get full lead detail & history
Calls	GET /api/v1/calls/	JWT Bearer	List all AI call records
Calls	GET /api/v1/calls/{id}/transcript/	JWT Bearer	View ElevenLabs turn-by-turn dialogue
Calls	GET /api/v1/calls/{id}/summary/	JWT Bearer	View AI Post-Call Qualification Summary
WhatsApp	GET /api/v1/whatsapp/	JWT Bearer	List WhatsApp message delivery logs
WhatsApp	POST /api/v1/whatsapp/{id}/resend/	JWT Bearer	Manual resend for failed WhatsApp message
Followup	GET /api/v1/followups/	JWT Bearer	List scheduled agent follow-ups
Metrics	GET /api/v1/dashboard/metrics/	JWT Bearer	Real-time CRM dashboard metrics
Telephony	POST /api/v1/telephony/webhooks/incoming-call/	Public	Exotel incoming call webhook
Telephony	POST /api/v1/telephony/webhooks/call-status/	Public	Exotel call status completion webhook

5. Plug-and-Play Live Provider Configuration (.env)

The backend is 100% plug-and-play ready. To activate live Exotel telephony, ElevenLabs AI voice, OpenAI LLM, and Meta WhatsApp Cloud API, configure the following keys in your .env file:

```
AI_CALLING_MOCK_PROVIDERS=false
```

```
EXOTEL_SID=your_exotel_sid
```

```
EXOTEL_API_KEY=your_exotel_key
```

```
EXOTEL_TOKEN=your_exotel_token
```

```
ELEVENLABS_API_KEY=your_elevenlabs_key
```

```
ELEVENLABS_AGENT_ID=your_elevenlabs_agent_id
```

OPENAI_API_KEY=your_openai_api_key

WHATSAPP_PHONE_NUMBER_ID=your_whatsapp_phone_id

WHATSAPP_ACCESS_TOKEN=your_whatsapp_token